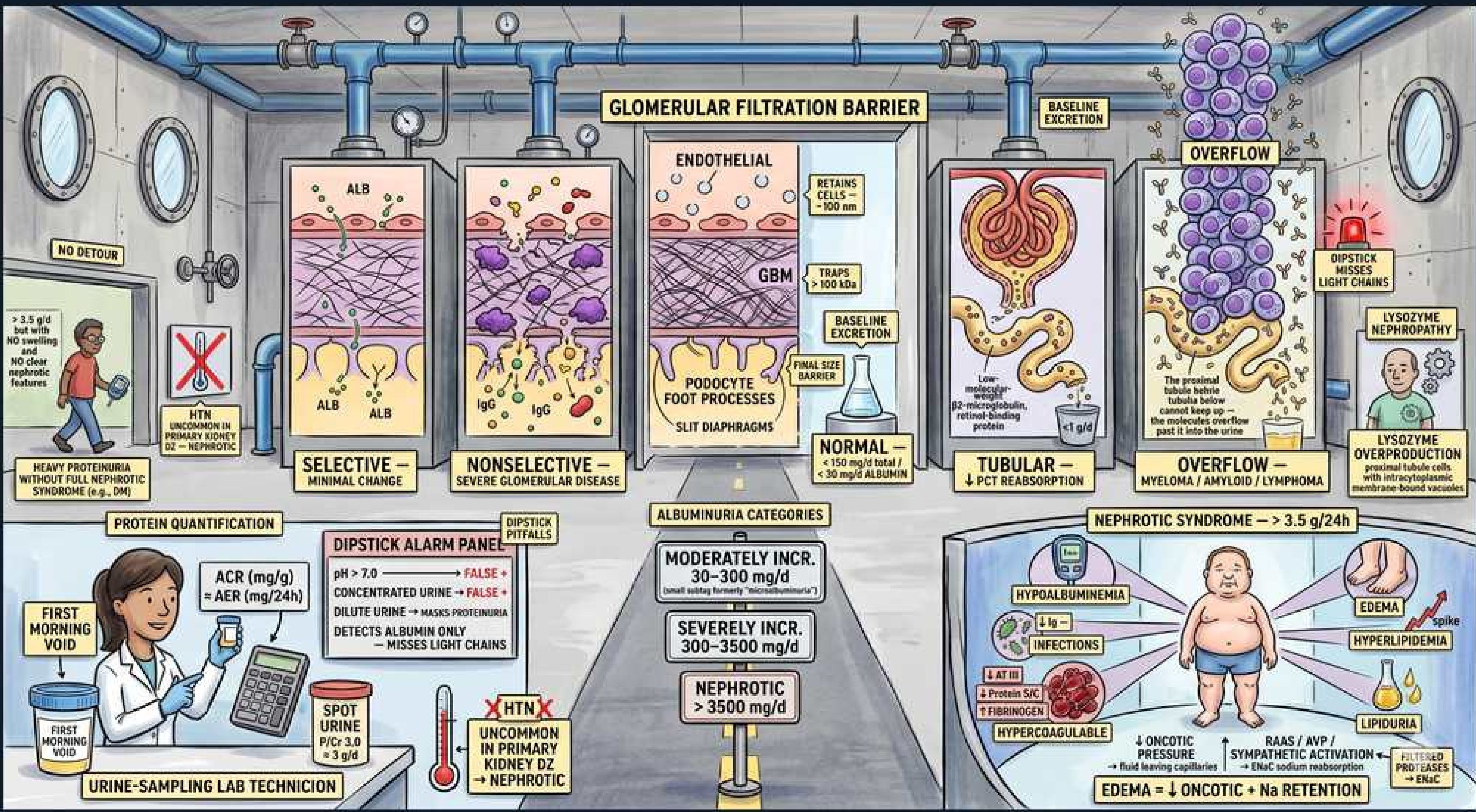
Renal Approach — Proteinuria (filtration barrier, glomerular/tubular/overflow, UACR, nephrotic)



1. Glomerular filtration barrier

WHAT YOU SEE

Top banner GLOMERULAR FILTRATION BARRIER over the layered wall panels

WHAT IT ENCODES

The filtration barrier has three layers: the fenestrated ENDOTHELIUM, the glomerular basement membrane (the main CHARGE barrier — heparan sulfate gives it a negative charge that repels anionic albumin), and the PODOCYTE foot processes with their slit diaphragms (the main SIZE barrier). Normally large/anionic proteins like albumin are excluded. Damage to charge selectivity leaks albumin; damage to size selectivity also leaks larger globulins (IgG). Mnemonic — 'endothelium, GBM (charge), podocyte slits (size).'

BOARD TRAP

Tempting wrong answer: 'minimal change disease causes structural destruction of the GBM seen on light microscopy.' Discriminator: minimal change is a CHARGE-barrier defect with effaced foot processes on ELECTRON microscopy but a NORMAL-appearing glomerulus on light microscopy — it produces selective (albumin) proteinuria, not the nonselective leak of severe structural disease.

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2. Selective proteinuria / minimal change

WHAT YOU SEE

Panel with ALB passing — SELECTIVE = MINIMAL CHANGE

WHAT IT ENCODES

Selective proteinuria means the urine protein is predominantly ALBUMIN with little larger globulin — the size barrier is intact and only the charge barrier is impaired. The prototype is minimal change disease (the most common nephrotic syndrome in children), which responds well to corticosteroids. Selectivity reflects a milder, more charge-specific glomerular lesion. Mnemonic — 'selective = small/single protein (albumin) = minimal change.'

BOARD TRAP

Tempting wrong answer: 'selective (albumin-only) nephrotic proteinuria indicates severe, steroid-resistant glomerular damage.' Discriminator: SELECTIVE proteinuria implies an intact size barrier and a milder lesion (minimal change) that is typically STEROID-RESPONSIVE; nonselective proteinuria (albumin + IgG) is what signals severe structural glomerular disease.

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3. Nonselective / severe glomerular

WHAT YOU SEE

Panel with ALB + IgG passing — NONSELECTIVE = SEVERE GLOMERULAR DISEASE

WHAT IT ENCODES

Nonselective proteinuria contains albumin PLUS larger proteins (IgG, transferrin) because both the charge and the SIZE barriers are breached — indicating more severe structural glomerular injury (e.g., focal segmental glomerulosclerosis, membranous nephropathy, diabetic/severe GN). It carries a worse prognosis and is less steroid-responsive than selective disease. Mnemonic — 'nonselective = big proteins escape = serious structural damage.'

BOARD TRAP

Tempting wrong answer: 'finding IgG in addition to albumin in the urine just means a more concentrated sample.' Discriminator: the presence of large proteins (IgG) is NONSELECTIVE proteinuria signaling a damaged SIZE barrier and severe glomerular disease — not a concentration artifact. It predicts worse outcome and points away from minimal change.

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4. Normal barrier excretion

WHAT YOU SEE

NORMAL panel <150 mg/d at barrier, baseline excretion

WHAT IT ENCODES

Normal total urinary protein is <150 mg/day, of which albumin is <30 mg/day; the rest is mostly secreted Tamm-Horsfall (uromodulin) protein and small filtered proteins that escape reabsorption. The healthy barrier excludes plasma proteins, and the proximal tubule reabsorbs the small amount that filters. Mnemonic — 'under 150 mg/day total, under 30 mg/day albumin = normal.'

BOARD TRAP

Tempting wrong answer: 'a total urine protein of 250 mg/day is within normal limits.' Discriminator: anything persistently above ~150 mg/day total protein (or >30 mg/day albumin) is abnormal and warrants evaluation — quantify and localize it rather than dismissing it as normal.

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5. Tubular proteinuria

WHAT YOU SEE

TUBULAR — low-molecular-weight proteins, ↓PCT reabsorption

WHAT IT ENCODES

Tubular proteinuria results when the PROXIMAL tubule fails to reabsorb the normally filtered LOW-molecular-weight proteins (β2-microglobulin, retinol-binding protein, α1-microglobulin). Causes are tubulointerstitial diseases — acute interstitial nephritis, Fanconi syndrome, heavy metals, drugs. The total amount is usually modest (typically <1-2 g/day) and dominated by LMW proteins, not albumin. Mnemonic — 'tubule can't reclaim the little proteins it normally recycles.'

BOARD TRAP

Tempting wrong answer: 'tubular proteinuria can reach nephrotic range and present with full nephrotic syndrome.' Discriminator: tubular proteinuria is usually MODEST (sub-nephrotic, LMW-protein predominant) and does NOT cause nephrotic syndrome; nephrotic-range (>3.5 g) heavy proteinuria is a GLOMERULAR process. The protein type (LMW vs albumin) and magnitude distinguish them.

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6. Overflow proteinuria

WHAT YOU SEE

OVERFLOW — myeloma / amyloid / lymphoma, light chains overwhelm tubules

WHAT IT ENCODES

Overflow proteinuria occurs when the plasma concentration of a small, freely filtered protein is so high that it overwhelms tubular reabsorption — classically monoclonal free light chains (Bence Jones protein) in multiple myeloma, also myoglobin and hemoglobin and lysozyme (monocytic leukemia). The barrier and tubule are intact; the problem is overproduction. Workup: serum/urine protein electrophoresis with immunofixation, serum free light chains. Mnemonic — 'overflow = too much small protein made, not a leaky kidney.'

BOARD TRAP

Tempting wrong answer: 'a normal/negative urine dipstick for protein rules out significant proteinuria in a patient with bone pain and anemia.' Discriminator: the dipstick detects ALBUMIN and MISSES light chains, so overflow proteinuria from myeloma can be dipstick-NEGATIVE despite heavy light-chain excretion. Order UPCR with urine protein electrophoresis/immunofixation and serum free light chains when myeloma is suspected.

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7. Dipstick misses light chains

WHAT YOU SEE

Right note: DIPSTICK MISSES LIGHT CHAINS, lysozyme

WHAT IT ENCODES

The urine dipstick uses a pH-sensitive dye that binds ALBUMIN, so it is insensitive to non-albumin proteins — monoclonal light chains, lysozyme, β2-microglobulin, and myoglobin can be heavily present yet read negative/trace. The classic clue is a dipstick-negative-but-high-total-protein discrepancy (sulfosalicylic acid test or UPCR detects total protein). Mnemonic — 'the stick only sees albumin; light chains slip past.'

BOARD TRAP

Tempting wrong answer: 'trace protein on dipstick excludes myeloma kidney.' Discriminator: a large gap between a low dipstick reading and a high quantitative total protein (UPCR) signals NON-albumin protein (light chains) — confirm with the sulfosalicylic acid test or electrophoresis. Never rule out myeloma cast nephropathy on the dipstick alone.

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8. Dipstick alarm panel

WHAT YOU SEE

DIPSTICK ALARM PANEL: pH >7.0, concentrated/dilute urine = FALSE positives/negatives

WHAT IT ENCODES

The dipstick is qualitative and concentration-dependent: highly ALKALINE urine (pH >7-8, e.g., contamination, UTI with urea-splitters, alkali) and very CONCENTRATED urine give FALSE-POSITIVE protein, while DILUTE urine gives FALSE-NEGATIVES. Gross hematuria, pus, semen, and antiseptics also confound it. Always confirm a positive dipstick with a quantitative ratio. Mnemonic — 'alkaline/concentrated = false +, dilute = false −.'

BOARD TRAP

Tempting wrong answer: 'a 2+ dipstick protein in a patient with very concentrated morning urine confirms pathologic proteinuria.' Discriminator: concentrated and alkaline urine cause FALSE-positive dipstick protein — confirm with a quantitative UACR/UPCR on an appropriate sample before labeling it pathologic.

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9. ACR / AER quantification

WHAT YOU SEE

Technician with calculator: ACR (mg/g) or AER (mg/24h), PROTEIN QUANTIFICATION

WHAT IT ENCODES

Quantify with the spot urine albumin-to-creatinine ratio (UACR, mg/g) or protein-to-creatinine ratio (UPCR) on a random sample — the creatinine in the denominator corrects for urine concentration, so the ratio approximates the 24-hour excretion (a UPCR of 3.5 ≈ 3.5 g/day) without a timed collection. The albumin excretion rate (AER, mg/24h) is the timed equivalent. Mnemonic — 'spot ratio ≈ grams per day, no jug needed.'

BOARD TRAP

Tempting wrong answer: 'a 24-hour urine collection is mandatory and superior for routine proteinuria monitoring.' Discriminator: spot UACR/UPCR is PREFERRED for screening and monitoring because timed 24h collections are error-prone (over/under-collection); the ratio corrects for concentration and tracks the 24h value reliably. Reserve 24h collections for select discrepant cases.

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10. First-morning / free void

WHAT YOU SEE

FIRST MORNING VOID / FREE WATER preferred sample

WHAT IT ENCODES

A first-morning-void specimen is preferred for quantifying proteinuria because it reflects protein excreted while the patient was recumbent overnight, minimizing diurnal and postural variation, and it specifically EXCLUDES orthostatic (postural) proteinuria. A normal first-morning sample after an abnormal random daytime sample points to benign orthostatic proteinuria. Mnemonic — 'morning void lies down the orthostatic noise.'

BOARD TRAP

Tempting wrong answer: 'proteinuria found on a random afternoon sample in a teenager confirms glomerular disease.' Discriminator: a normal FIRST-MORNING sample with protein only on upright daytime samples = benign ORTHOSTATIC proteinuria (common in adolescents/young adults, no treatment); don't pursue an aggressive glomerular workup until the morning sample is checked.

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11. Albuminuria categories

WHAT YOU SEE

Walkway: MODERATELY INCR 30-300, SEVERELY INCR 300-3000, NEPHROTIC >3500 mg/d

WHAT IT ENCODES

KDIGO albuminuria categories (UACR mg/g, ~mg/day): A1 normal-to-mildly increased <30; A2 MODERATELY increased 30-300 (old 'microalbuminuria' — earliest marker of diabetic nephropathy and a cardiovascular risk flag); A3 SEVERELY increased >300 (old 'macroalbuminuria'). These combine with the GFR (G) category to stage CKD risk. Mnemonic — '30 and 300 are the two cut-points: A1/A2/A3.'

BOARD TRAP

Tempting wrong answer: 'microalbuminuria of 30-300 mg/day is trivial and needs no action in a diabetic.' Discriminator: A2 'moderately increased' albuminuria is the EARLIEST sign of diabetic nephropathy and an independent cardiovascular risk marker — it mandates ACEi/ARB and risk-factor control, not reassurance. (Also note the terminology updated from micro/macro to moderately/severely increased.)

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12. Nephrotic range >3.5 g

WHAT YOU SEE

NEPHROTIC >3500 mg/d marker at the end of the walkway

WHAT IT ENCODES

Nephrotic-RANGE proteinuria is >3.5 g/day (or a UPCR >3.5 g/g), reflecting heavy glomerular protein loss. This is a quantitative threshold of protein only — it can exist WITHOUT the full clinical syndrome (e.g., early diabetic nephropathy, FSGS). Mnemonic — '3.5 grams marks the nephrotic doorway.'

BOARD TRAP

Tempting wrong answer: 'urine protein of 4 g/day = nephrotic SYNDROME.' Discriminator: nephrotic-range proteinuria (the number alone) is NOT the same as nephrotic SYNDROME, which additionally requires hypoalbuminemia, edema, and hyperlipidemia. A patient can have >3.5 g/day without the full syndrome — don't equate the two.

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13. Nephrotic syndrome features

WHAT YOU SEE

Edematous body: hypoalbuminemia, edema, hyperlipidemia, lipiduria, hypercoagulable, infections

WHAT IT ENCODES

Nephrotic SYNDROME = proteinuria >3.5 g/day + HYPOALBUMINEMIA (<3 g/dL) + EDEMA + HYPERLIPIDEMIA/lipiduria (oval fat bodies, 'maltese cross' under polarized light). Complications: hypercoagulability from urinary loss of antithrombin III (and protein C/S) → renal vein thrombosis/PE; infection risk from urinary immunoglobulin loss (encapsulated organisms, e.g., pneumococcal peritonitis); vitamin D and thyroid-binding protein loss. Mnemonic — 'PALE: Proteinuria, hypoAlbuminemia, hyperLipidemia, Edema.'

BOARD TRAP

Tempting wrong answer: 'a nephrotic patient with new flank pain and hematuria has a kidney stone.' Discriminator: nephrotic syndrome causes a hypercoagulable state from antithrombin III loss — new flank pain/hematuria or dyspnea should raise RENAL VEIN THROMBOSIS / PE (classically with membranous nephropathy), not a stone. Image with CT/Doppler and anticoagulate.

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14. Edema mechanism

WHAT YOU SEE

EDEMA = ↓ONCOTIC + Na RETENTION; RAAS/AVP/sympathetic activation

WHAT IT ENCODES

Nephrotic edema arises from two mechanisms: reduced plasma ONCOTIC pressure from hypoalbuminemia (favoring fluid into the interstitium) AND primary renal sodium retention (the 'overfill' mechanism — distal nephron Na avidity, with RAAS, AVP, and sympathetic activation). The relative contribution varies, but primary Na retention is often dominant. Treatment: dietary salt restriction and loop diuretics, plus RAAS blockade to reduce proteinuria. Mnemonic — 'low oncotic pulls fluid out; renal Na retention pours more in.'

BOARD TRAP

Tempting wrong answer: 'nephrotic edema is purely from low oncotic pressure, so albumin infusion is the definitive treatment.' Discriminator: primary renal SODIUM RETENTION is a major (often dominant) driver, so management centers on salt restriction, loop diuretics, and RAAS blockade — not albumin infusions, which give only transient benefit and don't address the renal Na avidity.
